

Comprehensive Security Analytics in the Internet of Things: An Exhaustive Analysis of the CICIoT2023 Benchmark, Historical Literature Evolution, and Unresolved Research Frontiers

Problem Statement Deep Dive

Fundamental Research Problem and Motivation

The pervasive integration of the Internet of Things (IoT) across industrial automation, smart healthcare, intelligent transportation, and smart home ecosystems has fundamentally restructured modern digital infrastructure¹. These distributed environments rely on dense networks of interconnected micro-controllers, sensors, and actuators that continuously capture, process, and exchange operational network traffic¹. However, this unprecedented connectivity has drastically enlarged the cyber-attack surface¹. Resource-constrained IoT endpoints are inherently vulnerable due to systemic architectural flaws, including unencrypted transmission channels, unpatched firmware vulnerabilities, lightweight protocol stack implementations, and weak default credentials¹. Consequently, compromised IoT nodes are frequently weaponized to form massive botnets capable of launching devastating Distributed Denial-of-Service (DDoS) campaigns, executing unauthorized surveillance, or facilitating lateral intrusion into enterprise networks¹.

Establishing effective Network Intrusion Detection Systems (NIDS) tailored specifically for IoT operations is a critical imperative for modern cybersecurity¹. However, the development, optimization, and empirical validation of Machine Learning (ML) and Deep Learning (DL) security analytics depend heavily on the availability of representative, high-fidelity, and rigorously annotated benchmark datasets¹. Historically, security researchers have relied on generic network traffic benchmarks, such as KDD1999, NSL-KDD, or CICIDS2017⁵. These datasets reflect traditional IT environments and fail to capture the unique traffic signatures, low-power communication behaviors, resource constraints, and multi-protocol ecosystems of modern IoT operations¹.

Critical Limitations of Prior IoT Security Datasets

To overcome the domain mismatch of legacy IT benchmarks, several IoT-specific attack datasets were proposed¹. However, a rigorous methodological review reveals four core

architectural limitations across this literature¹:

- **Topology Scale and Hardware Heterogeneity Constraints:** A significant majority of prior IoT benchmarks were constructed using highly restricted network topologies consisting of very few physical devices—frequently fewer than ten—or relying extensively on simulated or emulated nodes¹. These constrained environments fail to replicate the complex background noise, concurrent session density, and heterogeneous device interactions characteristic of real-world deployments¹.
- **Unrealistic Threat Models and Attack Execution Protocols:** In most existing datasets, malicious traffic is generated by external desktop computers, virtual machines, or dedicated servers targeting IoT victim devices¹. This design misrepresents real-world threat dynamics, wherein compromised, low-power IoT hardware acts as the primary vector to execute lateral, intra-network attacks against peer IoT endpoints¹.
- **Narrow Attack Vector Spectrum:** Legacy benchmarks predominantly focus on a narrow subset of attack types, heavily favoring high-volume botnet floods, such as Mirai and BASHLITE variants, or basic Denial-of-Service (DoS) vectors¹. Stealthy application-layer exploits, web vulnerabilities, multi-vector spoofing, dictionary brute-force attacks, and diverse packet fragmentation techniques are largely omitted¹.
- **Rigid Data Formats and Feature Provisioning:** Existing datasets often provide data either as raw binary packet captures without standardized tabular extraction tools, or as pre-processed flow summaries generated by static algorithms¹. This lack of standardization hinders flexible feature engineering and hampers reproducible comparative evaluation¹.

Non-Trivialities of IoT Intrusion Detection

Detecting malicious intrusions within IoT network traffic is fundamentally non-trivial due to severe operational and structural edge constraints¹:

- **High-Velocity Traffic vs. Resource-Constrained Hardware:** Edge gateways and IoT routers operate under strict processing, memory, and power limitations⁶. An effective NIDS must analyze high-velocity packet streams in real time without introducing processing latency or exhausting system resources¹.
- **Extreme Class Imbalance across Threat Classes:** Volumetric flooding vectors, such as ICMP or UDP floods, generate millions of malicious packets within seconds¹. Conversely, subtle application-layer attacks, such as Command Injection, Cross-Site Scripting (XSS), or dictionary brute-forcing, produce only a handful of packets¹. Standard supervised machine learning models trained on such imbalanced data inherently bias decision boundaries toward high-density classes, leading to severe misclassification of low-volume, high-risk intrusions¹.
- **Protocol Heterogeneity and Cross-Layer Behavior:** Modern smart topologies integrate local mesh protocols, such as Zigbee and Z-Wave, alongside transport-layer protocols like TCP/UDP and specialized application-layer standards like HTTP and MQTT¹. Learning generalized, high-discriminative behavioral profiles across these disparate

stacks requires robust multi-dimensional feature representations¹.

Stated Contributions of the CICIOT2023 Benchmark

To address these research gaps, the Canadian Institute for Cybersecurity (CIC) created the CICIOT2023 dataset¹. The paper presents several key advancements over prior literature¹:

- **Realistic High-Density Physical Topology:** The authors established a comprehensive smart-home testbed comprising 105 physical devices¹. This topology includes 67 direct network participants spanning smart home appliances, IP cameras, smart speakers, plugs, and micro-controllers, alongside 38 Zigbee and Z-Wave sensors managed via physical smart hubs¹.
- **Malicious IoT-to-IoT Attack Infrastructure:** The dataset captures authentic peer-to-peer threat dynamics by executing all malicious activities using 10 physical Raspberry Pi nodes acting as compromised IoT devices targeting vulnerable peer IoT endpoints¹.
- **Extensive Attack Diversity:** The benchmark incorporates 33 documented attack vectors categorized into 7 primary threat groups: DDoS, DoS, Reconnaissance, Web-based attacks, Brute Force, Spoofing, and Mirai botnet variations¹.
- **Dual Data Provisioning:** Data is published in both raw PCAP format (totaling 548 GB) and standardized tabular CSV format containing 46 statistical network features extracted over fixed packet windows of 10 and 100 packets¹.
- **Multi-Granular Machine Learning Evaluation:** The authors established performance baselines across three classification paradigms—binary (malicious vs. benign), 8-class threat category, and 34-class fine-grained attack classification—evaluating five supervised algorithms: Logistic Regression, Perceptron, AdaBoost, Random Forest, and Deep Neural Networks¹.

Literature Context via References

Historical Evolution Narrative of IoT Intrusion Benchmarks

The development of IoT intrusion detection research can be delineated into four sequential evolutionary phases¹. This progression highlights a gradual shift toward higher environmental realism, broader attack coverage, and more scalable analytical frameworks¹.

Phase 1: Early Anomaly and Botnet Scopes (2018)

Initial IoT intrusion detection research focused primarily on detecting specific botnet malware strains infecting small physical device setups⁸. Foundational benchmarks such as N-BaloT⁸ and Kitsune¹⁵ demonstrated that standard enterprise IT signatures were ineffective for resource-constrained IoT endpoints⁸. These studies introduced unsupervised deep autoencoders to establish baseline benign traffic profiles and flag deviations⁸. However, these early benchmarks were limited by small topology sizes (typically fewer than ten devices) and narrow threat models centered almost exclusively on Mirai and BASHLITE botnets¹.

Phase 2: Protocol-Specific and Synthetic Benchmarks (2019–2020)

To account for lightweight messaging protocols in smart environments, the second phase introduced protocol-specific datasets and multi-class classification frameworks⁹. Benchmarks such as BoT-IoT⁹ and MQTTset¹¹ evaluated traffic profiles across protocols like MQTT¹¹. However, datasets from this era suffered from structural limitations: BoT-IoT relied heavily on simulated and virtualized nodes, creating an extreme class imbalance where benign traffic comprised less than 0.01% of total network flows³.

Phase 3: Medium-Scale Hybrid and Telemetry Architectures (2020–2022)

Recognizing the drawbacks of fully simulated setups, researchers constructed medium-scale, hybrid environments⁵. Benchmarks like MedBloT¹⁰ integrated up to 83 real and emulated nodes to capture botnet infection, Command and Control (C2) communication, and attack execution lifecycles⁴. Concurrently, datasets such as Edge-IloTSet⁵ expanded benchmarking into Industrial IoT (IIoT) contexts, establishing performance baselines for centralized deep learning and privacy-preserving Federated Learning (FL) frameworks⁵.

Phase 4: Comprehensive Real-World IoT-to-IoT Benchmarking (2023–Present)

The release of CICIoT2023¹ marks the current state-of-the-art in IoT security benchmarking. By deploying 105 physical devices, orchestrating 33 distinct attack vectors, and ensuring that all malicious traffic originates directly from compromised peer IoT hardware, CICIoT2023 directly addresses the cumulative limitations of prior research phases¹.

Detailed Summaries of Key Reference Papers

N-BaloT (Meidan et al., 2018)

- **Approach and Methodology:** Proposed a network-based anomaly detection framework utilizing Deep Autoencoders (DAE) to extract behavioral snapshots from flow streams⁸. The framework collects 115 statistical features across 5 temporal window aggregations (100 ms, 500 ms, 1.5 s, 10 s, and 1 min) using the Argus tool¹⁶. Anomaly detection is performed by measuring the Reconstruction Error (RE) of network flows against a baseline trained exclusively on benign traffic¹⁷.
- **Testbed and Dataset:** Deployed 9 commercial IoT devices (e.g., smart doorbells, IP cameras, smart sockets, baby monitors) infected with real Mirai and BASHLITE botnet samples⁸.
- **Reported Results:** Achieved true positive rates (recall) between 0.9994 and 0.9997 with low false positive rates (0.0000 to 0.0071) across evaluated endpoints¹⁷.
- **Limitations:** Evaluated on a restricted setup of 9 devices¹; limited threat scope covering only two botnet families¹; severe imbalance between attack and benign traffic¹⁶; high computational overhead required for real-time extraction of 115 statistical features on resource-constrained edge hardware¹⁶.

BoT-IoT (Koroniotis et al., 2019)

- **Approach and Methodology:** Developed a botnet dataset designed for network forensic analytics and intrusion detection in IoT environments⁹. Employed statistical feature selection techniques (Pearson correlation coefficient and joint entropy) to identify optimal feature subsets, training Support Vector Machines (SVM), Random Forests, and Deep Neural Networks³.
- **Testbed and Dataset:** Built a hybrid virtualized testbed incorporating 5 target devices (e.g., Node-Red simulated nodes, weather stations, smart home hubs) generating over 72 million raw network records⁹. Covered DDoS, DoS, OS/Port Scans, Data Theft, and Keylogging vectors⁹.
- **Reported Results:** Achieved near-perfect binary classification accuracy (>99.99%) and multi-class accuracy ranging from 98.57% to 99.21% on selected feature subsets²⁰.
- **Limitations:** Heavy reliance on virtualized and simulated nodes¹; extreme class imbalance where benign traffic comprised less than 0.01% of total dataset records³; omitted application-layer web exploits, advanced spoofing, and diverse packet fragmentation vectors¹.

Kitsune (Mirsky et al., 2018)

- **Approach and Methodology:** Introduced KitNET, an online, unsupervised NIDS based on an ensemble of neural autoencoders¹⁵. Uses a feature mapper based on hierarchical clustering to route correlated packet features to small local autoencoders, enabling real-time anomaly detection directly on streaming packet data without offline training¹⁵.
- **Testbed and Dataset:** Evaluated across 9 physical IoT devices, including an IP camera infected with live Mirai botnet malware, alongside synthetic attack injections (e.g., ARP spoofing, SYN floods)¹⁵.
- **Reported Results:** Demonstrated high detection sensitivity and low execution latency, successfully flagging network anomalies prior to target endpoint disruption¹⁵.
- **Limitations:** Operates strictly as a binary anomaly detector; unable to categorize specific attack classes or identify fine-grained intrusion mechanisms²⁴; sensitive to feature grouping parameters and baseline traffic drift²⁴.

MedBloT (Guerra-Manzanares et al., 2020)

- **Approach and Methodology:** Designed a medium-scale behavioral IoT dataset focused on botnet infection, C2 communication, and attack execution lifecycles⁴. Extracted 100 statistical flow features evaluated using supervised classifiers (Random Forest, Decision Trees, LightGBM) and unsupervised anomaly detection models (One-Class k-NN, Isolation Forest)⁵.
- **Testbed and Dataset:** Deployed a medium-sized network topology comprising 83 total devices, combining physical commercial devices and emulated nodes⁴. Captured traffic from Mirai, BASHLITE, and Torii malware deployments⁴.

- **Reported Results:** Supervised classifiers (specifically LightGBM and Random Forest) achieved classification accuracy and macro F1-scores of 99.02% to 99.04%, while One-Class k-NN achieved 98% to 99% F1-scores in anomaly detection settings⁵.
- **Limitations:** Threat coverage largely restricted to botnet lifecycles and volumetric flooding attacks⁴; lacked web application vulnerabilities, ARP/DNS spoofing, and advanced protocol-specific exploits¹.

Edge-IIoTSet (Ferrag et al., 2022)

- **Approach and Methodology:** Created a cybersecurity benchmark tailored for Industrial IoT (IIoT) and smart environments⁵. Evaluated centralized machine learning models alongside Federated Learning (FL) frameworks to support privacy-preserving intrusion detection across distributed edge nodes⁵.
- **Testbed and Dataset:** Constructed a heterogeneous IIoT testbed integrating physical IoT sensors, PLCs, edge gateways, and cloud servers⁵. Included 14 attack vectors categorized into 5 clusters: DoS/DDoS, Information Gathering, Man-in-the-Middle, Injection, and Malware⁵.
- **Reported Results:** Centralized Deep Learning architectures achieved binary classification accuracy exceeding 99%, while Federated Learning configurations demonstrated strong cross-node detection performance⁵.
- **Limitations:** Primary focus oriented toward Industrial Control Systems (ICS/IIoT) topologies rather than high-density consumer smart environments⁵; attack execution pipelines relied on external attacking machines rather than intra-network compromised peer IoT hardware¹.

Comparative Landscape

Quantitative and Methodological Comparison Matrix

The following structured matrix compares the structural characteristics, methodology, topology scale, dataset composition, classification performance, and stated limitations of CIIoT2023 against its primary cited predecessors.

Dataset / Benchmark	Primary Analytical Approach	Network Topology & Hardware Scale	Attack Spectrum & Categories Covered	Key Classification Performance Metrics	Stated Primary Limitations
CIIoT2023 [cite: 1]	Supervised ML/DL (RF, DNN, LR,	105 physical devices (67	33 attack types in 7 categories	Binary Acc: 99.68% (RF) ¹	Macro F1-score degrades

	Perceptron, AdaBoost); DPKT window feature extraction ¹ .	direct direct nodes, 38 Zigbee/Z-Wave nodes; 10 Raspberry Pi attackers) ¹ .	(DDoS, DoS, Recon, Web, Brute Force, Spoofing, Mirai) ¹ .	8-Class Acc: 99.44% (RF) ¹ 34-Class Acc: 99.16% (RF) ¹ 34-Class F1: 0.714 (RF), 0.672 (DNN) ¹ .	significantly on low-volume , stealthy minority classes (Web, Brute Force) ¹ .
N-BalIoT (2018) [cite: 8]	Deep Autoencoders (DAE) for anomaly detection via Reconstruction Error; Argus extraction ⁸ .	9 commercial physical IoT devices (doorbells, cameras, webcams, smart plugs) ⁸ .	Mirai and BASHLITE botnet malware attacks ⁸ .	Recall: 0.9994–0.9997 ¹⁷ Precision: 0.9999–1.0000 ¹⁷ FPR: 0.0000–0.0071 ¹⁷ .	Small topology size (9 nodes) ¹ ; restricted to 2 botnet families ¹ ; high computational cost for 115 features ¹⁶ .
BoT-IoT (2019) [cite: 9]	Supervised ML/DL (SVM, RF, RNN, LSTM, CatBoost); Pearson correlation selection ³ .	5 target devices in a hybrid simulated/virtualized network environment ⁹ .	DDoS, DoS, OS/Port Scans, Data Theft, Keylogging ⁹ .	Binary Acc: >99.99% ²⁰ Multi-Class Acc: 98.57%–99.21% ²¹ F1-score: 0.984–0.998 ²² .	Virtualized testbed ¹ ; extreme class imbalance (<0.01% benign traffic) ³ ; lacks web/spoofing vectors ¹ .
Kitsune	Online	9 physical	Mirai	High online	Binary

(2018) [cite: 15]	unsupervised ensemble of Deep Autoencoders (KitNET) via feature mapping ¹⁵ .	IoT devices (IP cameras, video feeds, routers) ¹⁵ .	botnet, ARP spoofing, SYN flood, active scanning ¹⁵ .	detection rate; low anomaly latency prior to node failure ¹⁵ .	anomaly detection only ²⁴ ; cannot classify multi-class attack categories ²⁴ ; vulnerable to feature drift ²⁴ .
MedBloT (2020) [cite: 10]	Supervised ML (LightGBM, RF, DT) & Unsupervised Anomaly Detection (One-Class k-NN) ⁵ .	83 devices (combination of real physical and emulated IoT nodes) ⁴ .	Mirai, BASHLITE, Torii botnet propagation and C2 ⁴ .	LGBM Acc: 99.02% ²⁵ Threat Recall: 99.04% ²⁵ One-Class k-NN F1: 0.98–0.99 ⁵ .	Focused primarily on botnet lifecycles ⁴ ; missing broader web vulnerabilities and complex spoofing ¹ .
Edge-IIoTS et (2022) [cite: 5]	Centralized ML/DL (DNN, CNN, XGBoost) and Federated Learning (FL) frameworks ⁵ .	Heterogeneous IIoT testbed (sensors, PLCs, edge gateways, cloud servers) ⁵ .	14 attack types in 5 clusters (DoS, Scanning, MITM, Injection, Malware) ⁵ .	Centralized DL Acc: >99% ⁵ FL Macro F1: >0.92 ⁵ .	Tailored specifically for Industrial IoT (IIoT) ⁵ ; attacks driven by external entities rather than intra-IoT peer nodes ¹ .
MQTTset (2020)	Supervised ML	8 commercial	MQTT flood,	Binary Acc: 98.2%–99.4	Protocol-bound (MQTT

[cite: 11]	(Random Forest, Decision Trees, MLP) on MQTT protocol flows ¹¹ .	IoT devices connected to an MQTT broker ¹¹ .	authentication bypass, payload modification, DoS ¹¹ .	% ¹¹ Multi-Class F1: 0.91–0.96 ¹¹ .	only) ¹¹ ; fails to capture cross-layer transport and link-layer attack vectors ¹ .
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Methodological Contradictions and Disagreements in the Literature

Deep Learning vs. Classical Machine Learning Superiority

A major methodological debate across the reviewed literature centers on the relative performance of Deep Learning (DL) architectures versus classical Ensemble Machine Learning (ML) algorithms¹. Advocates of Deep Learning, such as the authors of N-BaloT⁸, Kitsune¹⁵, and recent spatial-temporal hybrid networks¹⁴, argue that deep layers automatically extract non-linear representations directly from streaming traffic, eliminating manual feature engineering bias⁸.

However, empirical findings from the CICIoT2023 benchmark¹, supported by independent comparative evaluations on N-BaloT²⁸ and ToN_IoT¹², explicitly challenge this assumption. In CICIoT2023, Random Forest consistently outperformed Deep Neural Networks (DNN) across binary, 8-class, and 34-class classification tasks¹. Specifically, Random Forest achieved a 34-class accuracy of 99.16% and a macro F1-score of 0.714, whereas the DNN achieved 98.61% accuracy and a macro F1-score of 0.672¹. Parallel studies confirm that classical algorithms like Random Forest and K-Nearest Neighbors (KNN) achieve higher accuracy (up to 99.95%) than deep CNN/RNN models (which averaged ~86.93% on complex multi-class subsets), while requiring orders of magnitude less computational memory and inference latency¹².

Feature Space Complexity vs. Edge Resource Constraints

The literature reflects conflicting strategies regarding optimal feature space dimensionality for IoT intrusion detection¹⁶. N-BaloT⁸ and its derivative frameworks¹⁶ advocate for high-dimensional feature spaces, relying on 115 statistical metrics computed across 5 temporal windows spanning 100 ms to 1 minute¹⁶. Proponents contend that multi-window tracking is necessary to distinguish subtle botnet communication from legitimate user activity¹⁶.

Conversely, lightweight optimization studies demonstrate that applying dimensionality reduction (e.g., PCA, Boruta, or Information Gain filtering) to shrink feature spaces down to 5 to 10 core parameters yields near-identical classification accuracy (>99%) while dramatically lowering memory footprints and processing time¹². CICIoT2023 adopts a balanced approach by providing 46 statistical features extracted over small packet windows (10 and 100 packets)¹. This demonstrates that localized, window-aggregated metrics achieve robust multi-class

separation without the overhead of long-term temporal tracking¹.

Synthetic Oversampling vs. Natural Distribution Preservation

Researchers disagree sharply on how to address the severe class imbalance inherent in network intrusion data¹. Studies extending the BoT-IoT dataset rely on synthetic data generation techniques—such as SMOTE (Synthetic Minority Over-sampling Technique) or WGAN-GP (Wasserstein Generative Adversarial Networks)—to artificially balance rare attack or benign classes, reporting perfect detection rates post-oversampling³.

Conversely, the authors of CICIoT2023¹ and N-BalIoT¹⁶ caution against synthetic oversampling in network security contexts. They argue that algorithms like SMOTE generate synthetic feature points in non-Euclidean space, creating unrealistic flow combinations that fail to respect underlying packet transmission mechanics¹⁶. Instead, CICIoT2023 preserves natural traffic distributions during packet capture and addresses imbalance through window-based averaging and stratified subsampling during model training¹.

Open Problems and Capstone Scope Justification

Unresolved Research Gaps in IoT Intrusion Detection

Macro F1-Score Collapse on Low-Volume Stealthy Attack Classes

While state-of-the-art classifiers evaluated on CICIoT2023 achieve overall classification accuracy exceeding 99%, granular class-level evaluation reveals a substantial performance drop when moving from binary to fine-grained multi-class classification¹.

In the 34-class evaluation, Random Forest achieves an overall accuracy of 99.16%, but its macro-averaged F1-score falls to 0.714; similarly, the Deep Neural Network achieves 98.61% accuracy but a macro F1-score of only 0.672¹. As shown in the CICIoT2023 confusion matrices, volumetric flooding attacks (such as DDoS ICMP, UDP floods, and Mirai variants) are classified with near-perfect accuracy¹. However, stealthy application-layer threats—including SQL Injection, Command Injection, Cross-Site Scripting (XSS), Uploading Attacks, and Dictionary Brute Force—are frequently misclassified as benign traffic, Reconnaissance, or Spoofing¹. Because stealthy attack vectors contribute only a minute fraction of total dataset records (e.g., 1,252 uploading attack samples versus 7,200,504 DDoS ICMP samples)¹, standard loss functions (such as Cross-Entropy) are dominated by majority-class gradient updates¹. Developing specialized cost-sensitive loss functions, focal loss formulations, or hierarchical class-aware neural decoders capable of preserving high sensitivity on low-volume, high-risk attack vectors remains a key open challenge².

Cross-Domain Feature Transferability and Distribution Shift

A core limitation across contemporary IoT security literature is the reliance on single-dataset evaluations¹³. Models trained and validated on a specific benchmark (e.g., CICIoT2023) experience noticeable performance degradation when deployed directly onto unseen IoT

topologies or evaluated against external datasets (e.g., N-BaloT, ToN_IoT)¹². This degradation stems from multiple factors¹¹:

- Differences in network hardware configurations, topology density, and background traffic dynamics¹¹.
- Discrepancies in feature extraction tools (e.g., DPKT¹ versus Argus¹⁶ versus CICFlowMeter⁷ versus NFStream¹).
- Concept drift and evolving attack signatures across operational environments¹².

Developing domain-adaptation techniques, transfer-learning pipelines, and tool-agnostic feature representations that generalize across heterogeneous IoT environments represents a crucial unresolved problem¹³.

Real-Time Edge Processing Overhead vs. Streaming Feature Extraction

Existing intrusion detection pipelines typically separate packet capture from feature extraction and model inference¹. In live production settings, computing window-based aggregations (e.g., magnitude, covariance, directional variance over 100-packet windows) introduces significant memory buffer management and processing overhead¹. During high-velocity DDoS flooding, edge gateways can experience resource exhaustion within the feature extraction pipeline before the classification model even executes¹³. Research into end-to-end raw packet deep learning architectures or hardware-accelerated feature extraction primitives remains an urgent priority²⁸.

Explainability, Interpretability, and Adversarial Resilience

Despite the high predictive metrics achieved by Deep Neural Networks and Random Forests, these models operate largely as black boxes³². In operational Security Operations Centers (SOCs), analysts require interpretable explanations detailing *why* a specific sequence of network traffic was flagged as malicious¹². Furthermore, existing ML and DL models remain vulnerable to adversarial evasion attacks, where malicious actors subtly alter packet inter-arrival times (IAT), payload padding, or packet length distributions to bypass detection models without disrupting the underlying exploit¹². Designing explainable AI (XAI) frameworks that remain robust against adversarial traffic manipulation is an important open frontier in IoT defense¹².

Strategic Capstone Project Scope and Contribution Justification

The gaps identified in the CICIoT2023 benchmark and its broader literature base provide a solid foundation for defining a capstone project. A student can structure their capstone around one of the following high-impact research directions:

- **Addressing Class Imbalance via Class-Aware Loss Engineering:**
 - *Objective:* Design and evaluate a neural network architecture incorporating Focal Loss, Cost-Sensitive Resampling, or a Semantic Class-Aware Layered Decoder (SCALED)² specifically tuned to address the macro F1-score drop on low-volume attack classes in CICIoT2023 (e.g., Web-based attacks and Brute Force) without

degrading volumetric DDoS detection accuracy¹.

- **Evaluating Cross-Dataset Generalizability and Transfer Learning:**
 - *Objective:* Construct a domain-adaptation pipeline that extracts a standardized feature set across CICIoT2023, N-BaloT, and Edge-IIoTSet¹. Evaluate cross-dataset transferability to quantify and mitigate model performance degradation caused by topology shifts and hardware heterogeneity¹³.
- **Ultra-Lightweight Edge Feature Optimization for Real-Time Inference:**
 - *Objective:* Perform a detailed feature-importance and ablation study on the 46 features in CICIoT2023 using SHAP (SHapley Additive exPlanations) and Boruta selection algorithms¹². Implement a low-dimensional feature extractor capable of sub-millisecond execution on resource-constrained hardware, such as a Raspberry Pi or NVIDIA Jetson edge node²⁸.
- **Adversarial Robustness Evaluation of Machine Learning NIDS:**
 - *Objective:* Build an adversarial traffic modification pipeline that applies packet perturbations (such as altering inter-arrival times and packet size distributions) to CICIoT2023 attack streams¹². Benchmark the performance degradation of standard ML classifiers and propose an adversarial training defense mechanism to improve detection robustness in IoT networks¹².

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